Clustering of Queries Using Geometric and Semantic Features

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***Abstract*—Clustering search queries has an important value in e-commerce systems as it helps to understand users intent , and this improves many related things like recommendation systems , so in this study we used various clustering algorithms on amazon queries dataset , and the clustering was based on three different representation first on only geometric of queries (number of words , chars , digit , special characters..) , then only semantic embeddings , and finally on both geometric and semantic embeddings , different clustering methods were compared using internal evaluation metrics and qualitative observation of clusters , and as result based on only geometric it gives simple clusters (like cluster 0 queries with no detailed info with no numbers , cluster 1 queries with much numbers and symbols) , it generate a well structured clusters, while semantic gives clusters with more meaning product categories , and for geometric + sematic it gives a complex clustering space .**

keywords–Queries Clustering, Unsupervised Learning, Semantic Embeddings, Geometric of Queries, KMeans, GMM, DBSCAN, HDBSCAN, Dimensionality Reduction, Amazon Queries Dataset

I. INTRODUCTION

The fast growth of e-commerce systems leads to a high volume of search and user generated queries, for this understanding what users really wants it is very important especially for big platforms like amazon [1], clustering those queries it's an effective way to uncover the hidden patterns in users behavior , and based on this patterns we could improve User experience, search results , recommendation systems…

Query clustering aims to group similar queries together without relaying on labeled data , traditional approaches often rely on just geometric features such as query length , number of words , specific patterns , this structural features can provide some good clusters [4] , but they fail at capturing the meaning which is very important to understand user intent.

But now with advanced technique we can give a mathematical meaning for this queries , through transformer based model [8] , that have enabled the extraction of semantic embeddings , these embeddings allow similar queries to be close to each other in high dimensional space [3].

In this study we investigates the clustering of Amazon search queries using multiple algorithms (Kmeans , GMM , DBSCAN , HDBSCAN) [5] and we did compared main settings clustering based on traditional features only Geometric , and on semantic only , and based on combination of geometric and semantic , the evaluation made by metrics like Silhouette score , Calinski-Harabasz index and Davies-Bouldin index , and also with observation or qualitative analysis of cluster coherence.

II. STATE OF THE ART

Many recent studies worked on clustering text and search queries using semantic features and unsupervised learning algorithms . traditional approaches mainly used statistical or geometric text features , and recent works use transformer based models such as Sentence-BERT to better capture semantic meaning of queries [3].

Aggarwal and Zhai [4] showed that the combination of structural and semantic features in text mining tasks can improve the results, but also increase the complexity if the two feature spaces are not compatible. This corroborates our observation in the hybrid experiment.

Among the most used clustering algorithms, KMeans is simple and fast but sensitive to cluster shape and initialization [14]. GMM is more flexible but can be affected by overlapping data [10]. DBSCAN and HDBSCAN can detect clusters with different shapes and identify noise points, but their performance depends on density distribution and parameter selection [5].

Dimensionality reduction methods such as PCA and UMAP are also commonly used to reduce embedding dimensions and improve clustering visualization [2].

The use of semantic embeddings in our work was mainly motivated by the Sentence-BERT model proposed by Reimers and Gurevych [3], which showed that transformer based embeddings lead to a better semantic sentence similarity compared to traditional methods. This was made possible by the Transformer architecture proposed by Vaswani et al. [8].

McInnes et al. [5] showed that HDBSCAN handles datasets with different density better than DBSCAN, which is of importance when working with real user queries with very different

lengths and structures. Rousseeuw [17] also mentioned that the Silhouette score is not always a good enough measure of cluster quality, which is also a reason why we use several metrics along with qualitative observation.

III. BACKGROUND

A. Dataset

This work was applied on Amazon shopping queries dataset examples [18] , a dataset that was principally designed to be combined with another dataset of products info in order to cluster the matching of queries and products , which could eventually results in 4 clusters or labels ESCI.

This dataset contains real different search queries , and it Contains queries in English , Japanese and Spanish. Our objective was not following the same purpose of this dataset , but to cluster only the query by it self.

TABLE I
AMAZON ESCI DATASET

DATASET	Number of rows	Number of features
Amazon ESCI Queries	2,621,288	9 ('example_id', 'query', 'product_locale'…)

B. Clustering Models

a) K-Means: An Unsupervised ML algorithm , it groups similar data point into clusters without the need of labeled data [14] , the number k of clusters is predefined , it assumes that clusters are spherical and with similar size , which makes it efficient but sensitive to outliers .

b) Gaussian Mixture Models (GMM): It's a probabilistic clustering technique, unlike K-Means GMM provides probabilistic cluster assignments [10], this feature allows a data point to belong to multiple clusters with different probabilities.

c) Density-Based Spatial Clustering of Applications with Noise (DBSCAN): A density-based clustering algorithm that groups together data points that are close, and defines the outliers as noise [15], unlike previous algorithms DBSCAN does not require a specific number of clusters in advance, its performance totally depending on the choice of density parameters.

d) Hierarchical DBSCAN (HDBSCAN): Is an extension of DBSCAN that builds a hierarchy of clusters and selects the most stable ones [5], and it can handle better datasets with varying density. And also it can identify noise points making it good with datasets with outliers.

IV. METHODOLOGY

A. Metrics, Methods and Models

We used some metrics , methods for evaluation , analysing and also a pretrained model to give a meaning for queries.

a) Silhouette Score: A score that measures how similar data point is to its own cluster compared to others , the score ranges from -1 to 1 where higher values indicate better, defined and more separated clusters [17] . The mathematical representation is :

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

b) Calinski-Harabasz index: Evaluates the ratio between inter cluster dispersion and intra cluster dispersion , a higher value indicates that clusters are well separated [9] , this metric tends to favor solutions with a large number of clusters,The mathematical representation is :

$$CH = \frac{SS_B}{SS_W} \times \frac{N-k}{k-1}$$

c) Davies Bouldin Inedx: Measures the average similarity between each cluster and its most similar one , and considers cluster spread and distance between clusters , lower value indicate better clustering [12] ,The mathematical representation is :

$$DB = \frac{1}{k} \sum_{i=1}^{k} \max_{j \neq i} \left(\frac{\sigma_i + \sigma_j}{d(c_i, c_j)} \right)$$

d) principal component analysis (PCA): An algorithm with the objective of reducing dimensionality of the data , and to simplify visualization in 2D [7] ,The mathematical representation is :

$$\Sigma = \frac{1}{N} \sum_{i=1}^{N} (x_i - \mu)(x_i - \mu)^T, \quad \Sigma \mathbf{v} = \lambda \mathbf{v}$$

e) t-SNE: is a technique of visualisation used to project high dimensional into 2d spaces , unlike pca , t-sne focuses more on preserving local stracture and neighborhood relations between points [6] The mathematical representation is :

$$q_{ij} = \frac{(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1}}{\sum_k \sum_{l \neq k} (1 + \|\mathbf{y}_k - \mathbf{y}_l\|^2)^{-1}}$$

f) Uniform Manifold Approximation and Projection (UMAP): is also another dimensionality reduction technique used to transform high dimensional data into smaller representation while preserving important structures of the data [2] . In this work, UMAP used to reduce dimensionality of semantic embeddings before clustering. This helped reduce computation time and memory usage while keeping most semantic relationships between queries. Compared to PCA, UMAP is often better at preserving local similarities in embedding spaces [2] , the mathematical representation is :

$$w(a,b) = \exp\left(\frac{-(d(a,b) - \rho_a)}{\sigma_a}\right)$$

g) SentanceTransformer (all-MiniLM-L6-v2): To capture the meaning of queries a semantic representation was generated using a transformerbased model from Sentence-Transformers library [3]. Specifically, the pre-trained model "all-MiniLM-L6-v2" was used to convert each query into a 384D dense numerical vector , where the semantic properties are encoded as latent features in a high-dimensional space.

1. Vector Encoding : Each query "Q" is converted into a vector .This is achieved by taking the hidden states of all n tokens in the query and applying Mean Pooling: [3]

$$\mathbf{u} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{h}_i$$

2. Semantic Similarity for Clustering : To facilitate clustering,

the model ensures that queries with similar meanings are mathematically close. The "closeness" between two query vectors, u and v is measured using Cosine Similarity : [3]

$$\text{Similarity}(\mathbf{u},\mathbf{v}) = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|} = \frac{\sum_{m=1}^{d} u_m v_m}{\sqrt{\sum_{m=1}^{d} u_m^2}\sqrt{\sum_{m=1}^{d} v_m^2}}$$

B. Preprocessing

Before applying clustering algorithms , we preprocessed the amazon queries dataset , the goal was to clean data and extract useful features , and prepare the queries for clustering , both geometric features and semantic embeddings are generated from the original query text.

At first we removed all columns except one "query" that's our source of other columns , from this column we extracted "query_chars" , "query_words" , "contains_digit" with 0 and 1 values , "digit_count" , "avg_word_len" calculated by number of chars devided by number of words , "space_count" , "spec_char_count" number of special characters it could help detect a pattern like question queries , or technique queries , and for number of words and chars could help detect clusters of like detailed queries or direct queries .

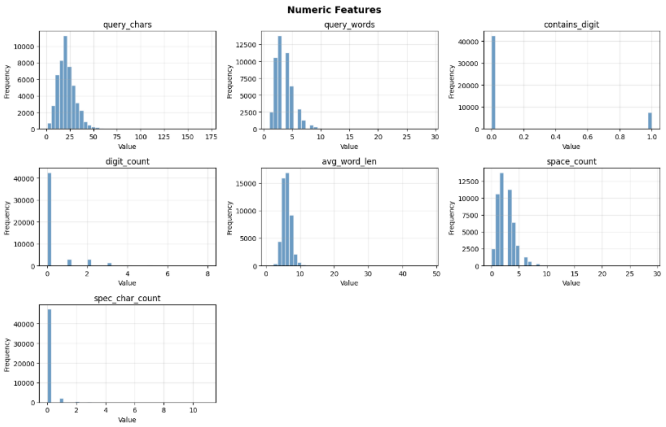


Fig. 1. Geometric Features distribution

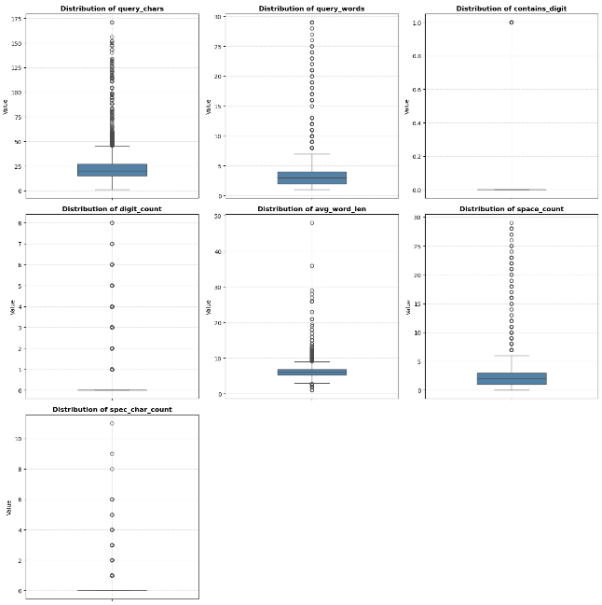


Fig. 2. Geometric Features BoxPlot

This geometric features have many outliers , and that's normal since we are dealing with real queries , there is users that write normal queries , and there is users that write very big queries , and we choose to keep them , we have to cluster also this type.

And we scale all those columns it's very important for the algorithms that we used we used standardization to put all features in one scale :

$$z = \frac{x - \mu}{\sigma}$$

And one important thing is we choose US queries to simplify and to understand the clusters , and for null values none since we generated all columns from the query column (that it is also having no null values)

For the other part we added the meaning of those queries using a sentenceTransformer Model that gives 384 columns or features , and to simplify clustering we applied UMAP to minimize the number of this features to 15 we choose it as a compromise between preserving semantic meaning and reducing computational complexity.

UMAP was used for dimensionality reduction because it works well with high dimensional embedding data and keeps semantic relationships between queries better than many traditional methods [2].

while PCA mainly keeps global variance , UMAP focuse more on keeping nearby points close to each other [2] , and after we combine all into one dataset of 50,000 random rows to reducing computational time.

TABLE II
AFTER PREPROCESSING DATASET

Cleaned DATASET	**Number of rows**	**Number of features**
Cleaned Amazon English Queries	50,000	23 (query + 7 geometric + 15 Semantic ...))

C. Parameter Selection

We used some methods and approaches , to find better parameters on the algorithms we used in clustering .

a) KMeans (K parameter):

To choose the best number of clusters at each case (geometric / semantic / hybrid) for KMeans we used 2 methods Elbow Method [11] and Silhouette score at each K [13] , the elbow curves did not show a very clear optimal in all cases , so we based on the second method , we choose K based on Silhouette score like (semantic case : K=80 Silhouette score=0.37) .

b) DBSCAN (eps parameter):

For DBSCAN the K-distance graph was used to get a suitable value for epsilon parameter [16] , the selected value indicate point where distances start increasing more rapidly.

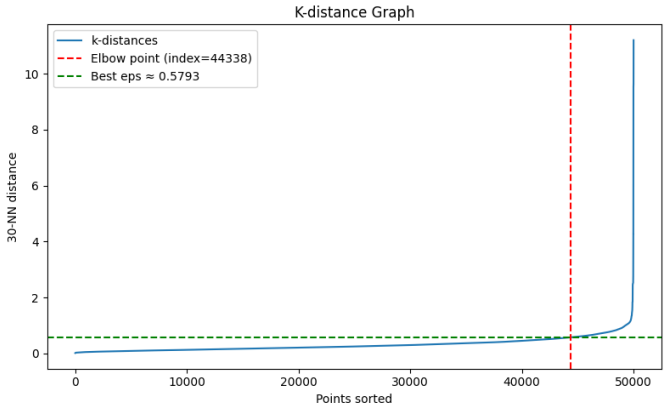


Fig. 3. K-distance to find best eps for DBSCAN – semantic case

c) HDBSCAN (min_cluster_size, min_samples, and cluster_selection_epsilon parameters):

For HDBSCAN different parameters combination were tested using an optimization approach, parameters such as min_cluster_size, min_samples, and cluster_selection_epsilon were tuned in order to maximize clustering quality according to internal evaluation metrics.

V. FINDINGS AND ANALYSIS

A. Influence of Preprocessing

Scaling and reduction of dimensionality had significant influence on clustering results. Since clustering algorithms that we use Kmeans GMM DBSCAN HDBSCAN are based on distance calculations , features with larger values can dominate the clustering process if scaling is not applied as we experiment on geometric only.

In geometric features such as number of words or characters count had more influence than smaller features like is contain_digit (0 or 1) , as we can see in graph bellow it's clustering based on geometric features with no scaling , we used Kmeans K=2 , it clusters into 2 groups big queries and smaller ones , it ignores all other features and focused only on the features with bigger values .

After applying scaling (next experiments) , the contribution of features became more balanced, which produced more stable clusters , and the same goes for dimensionality reduction in our case and specially when we cluster based on semantic features (384D) the algorithms are so slow so dimensionality reduction it's also important when we have large number of features , This shows that preprocessing and scaling are important steps in clustering.

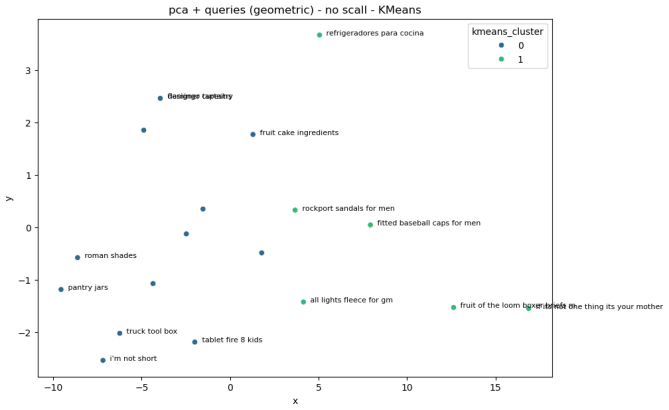


Fig. 4. A representation of queries using pca without scalling – geometric case

B. Clustering based on only geometric features

In this first experiment clustering was performed using only geometric features extracted from queries, the quantitative results of clustering summarized in this table:

TABLE III
CLUSTERING RESULTS - GEOMETRIC FEATURES ONLY

Model	Silhouette score	Calinski Harabasz	Davies Bouldin	Clusters Number
KMeans	0.53	20764.16	1.13	2
GMM	0.51	18616.37	1.26	2
DBSCAN	0.84	15745.92	0.37	91
HDBSCAN	0.91	19067.76	0.48	163

By analyzing this numbers we can see that all algorithms get relatively high Silhouette scores , especially for DBSCAN , and HDBSCAN , and a higher Silhouette score means data points are closer to their own cluster and far from others , while the Calinski Harabasz index is also high , means clusters are well compact and well separated globally.

Number of clusters is also important when interpreting these metrics. DBSCAN and HDBSCAN have a very large number of clusters, which can increase metric artificially because smaller clusters are easier to separate , in the case of KMeans, the Elbow Method did not show a very clear optimal point. And we used also Silhouette score at each K (number of clusters) , the best value was obtained for K = 2 (0.53) , while larger values of K sometimes produced even higher scores after more than 100 clusters , and we could choose also a higher K for Kmeans.

But Higher metric values do not always means better or more meaningful clustering , in this first case, geometric features mainly separate queries based on structure, such as query length or presence of numbers, which creates mathematically clean clusters but not necessarily useful.

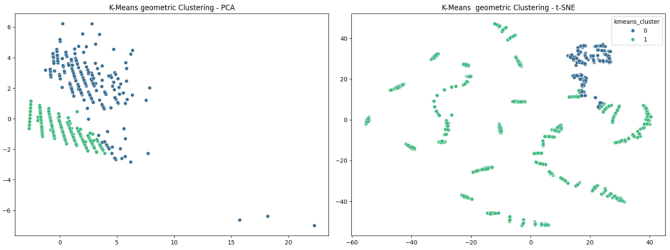


Fig. 5. A representation of Kmeans Clustering using PCA and t-SNE - geometric case

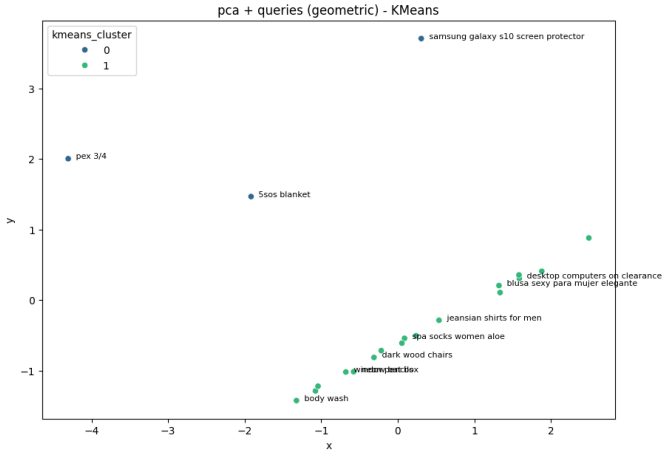


Fig. 6. A representation of queries using pca KMeans – geometric case

For example Kmeans and GMM at K=2 they cluster queries into 2 groups , a group that contains queries with numbers and queries with no numbers and GMM shows a high confidence at clustering in 2 groups.

And based on PCA graph we can see a clear 2 groups , and we can say that PCA looks at the big picture while t-SNE it looks more detailed and we can see there is like 6-8 groups and if we use Kmeas at K=6 it will generate sub clusters like clusters with number of words , big queries and smaller ones.

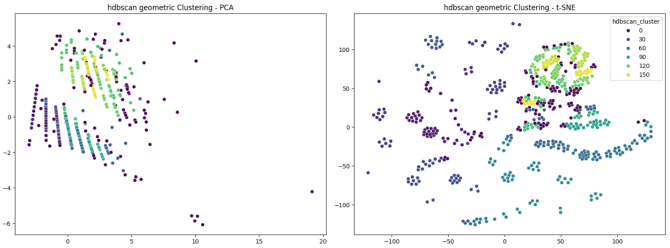


Fig. 7. A representation of HDBSCAN Clustering using PCA and t-SNE - geometric case

For DBSCAN , HDBSCAN since we don't choose K number they generated many clusters, like very specified categories (cluster with 2 words , cluster with 3 words ...) .

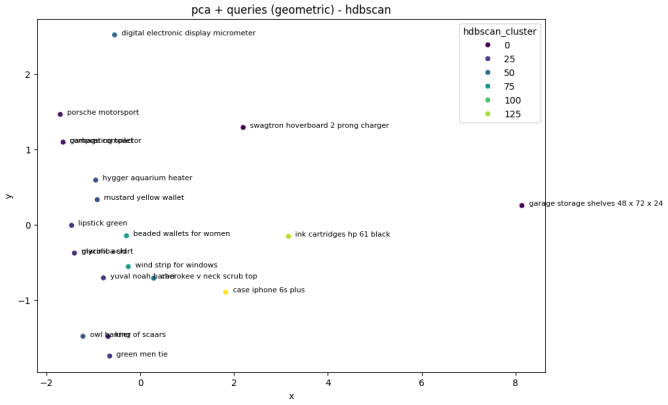


Fig. 8. A representation of queries using PCA - HDBSCAN – geometric case

and they both define as a noise the queries with like symbols and no spaces or no clear queries cluster -1 : ‘jewellery+box+small+crystal’, ‘hose reel 1/4’, ‘bat181’ .

Overall, it is difficult to determine a single best clustering solution based only on internal metrics. Qualitative observation of cluster examples is also necessary to understand whether the clusters are truly meaningful.

C. Clustering based on only Semantic Features

And in this experiment clustering was performed using only semantic embeddings features, and the quantitative results of clustering summarized in this table:

TABLE IV
CLUSTERING RESULTS - SEMANTIC FEATURES ONLY

Model	Silhouette score	Calinski Harabasz	Davies Bouldin	Clusters Number
KMeans	0.36	9247.88	1.04	80
GMM	0.32	7907.38	1.13	85
DBSCAN	0.22	2527.68	0.85	134
HDBSCAN	0.38	7240.16	0.93	91

Compared to geometric experiment Silhouette and Calinski Harabasz scores became lower for most models , meaning semantic clusters are less clearly separated.

KMeans and HDBSCAN achieved the best overall balance in this experiment. HDBSCAN had the highest Silhouette score , while KMeans produced similar or stable results plus a large number of clusters. DBSCAN generated many clusters but with weaker separation compared to geometric case.

Davies Bouldin scores were generally better indicating that semantic clustering still produces reasonably distinct groups, even if the clustering metrics are generally lower than geometric case but in terms of observation of queries , we could find more meaning clusters , clusters related to similar products or categories.

Silhouette score is low for DBSCAN since its often does not work well in the lower dimensional space produced by UMAP due to its dependence on an inflexible global epsilon that is not adaptable to the varying densities of high dimensional text embeddings, leading to over-chattering of clusters or even marking genuine semantically coherent clusters as outliers, as stated by McInnes and Healy. [19]

Another important observation is that semantic clustering generated a larger number of clusters compared to geometric case, this suggests that semantic embeddings capture more detailed information between queries.

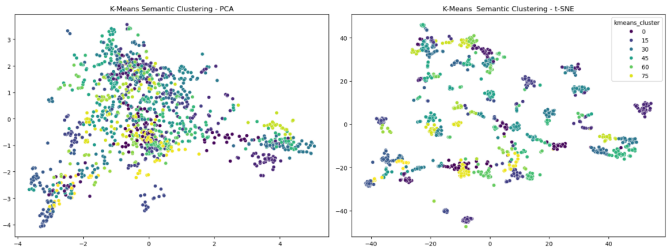


Fig. 9. A representation of Kmeans Clustering using PCA and t-SNE - Semantic case

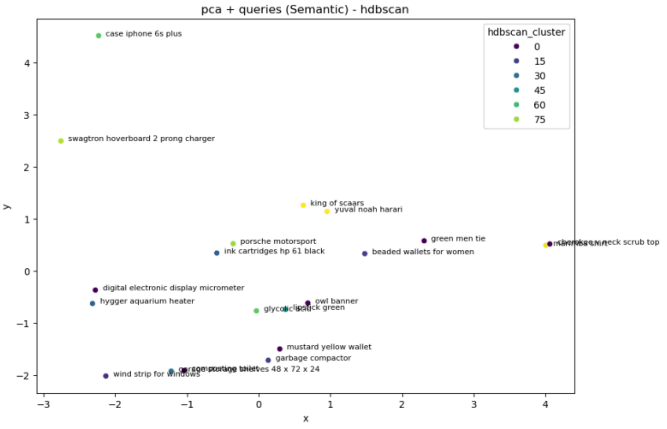


Fig. 10. A representation of queries using PCA - HDBSCAN – Semantic case

All algorithms gives a similar clustering in terms of observation many groups each group contain sub categories of products (because they are real queries of amazon) , and in terms of visualization pca no clear groups because of the complexity of semantic clusters , and T-sne we can see like 14 number of clusters .

Overall, semantic clustering gives lower mathematical separation than geometric clustering, but it produces clusters that are more useful for understanding user intent and query meaning.

D. Clustering on both Semantic and Geometric Features

Now clustering was performed using geometric features and semantic embeddings , and as a note we scale both geometric and semantic to be in the same scale after noticing a slight

influence of semantic features , and the quantitative results of clustering summarized in this table:

TABLE V
CLUSTERING RESULTS - HYBRID GEOMETRIC AND SEMANTIC FEATURES

Model	Silhouette score	Calinski Harabasz	Davies Bouldin	Clusters Number
KMeans	0.179	2345.78	1.53	65
GMM	0.125	1640.37	2.19	65
DBSCAN	-0.022	567.17	1.45	10
HDBSCAN	0.022	643.39	2.08	108

In this experiment the results shows that all models have lower scores compared to other cases, the Silhouette score are much smaller in all algorithms , and for Davies Bouldin are much higher means clusters are not well separated , and there is a lot of overlap between clusters.

DBSCAN gives a negative Silhouette score , which means many points are closer to other clusters than their own , this because DBSCAN has a problem it uses a fixed epsilon value , this value does not work well with the scales of spatial and semantic manifolds as what we already discovered in semantic case this makes DBSCAN unstable [19] , and HDBSCAN gives a very small Silhouette score , and a big number of clusters.

This topological incompatibility causes this profoundly unbalanced and failure profile in cluster assignment from DBSCAN, a massive central cluster that occupies the space, and others that are fragmented or rejected. as it shown in the graph below

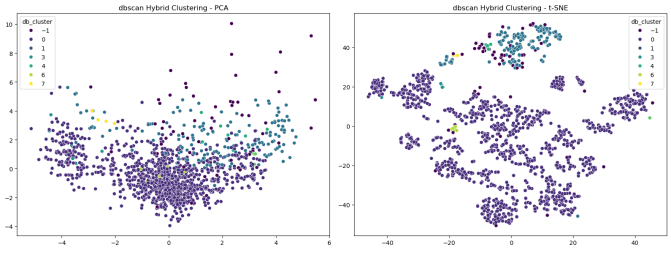


Fig. 11. A representation of DBSCAN Clustering using PCA and t-SNE - geometric + semantic case

This happens because this 2 types of features geometric and semantic describing 2 different aspect , geometric focuses on structure while semantic on meaning , which means a lack of topological compatibility between the geometric and semantic manifolds, which results in an erroneous distance function and the ”shattering” of the cluster structure [4] .

But in terms of observation we could find useful patterns in clusters like Kmeans it produced many clusters sometimes combining between product category + number of words for example:

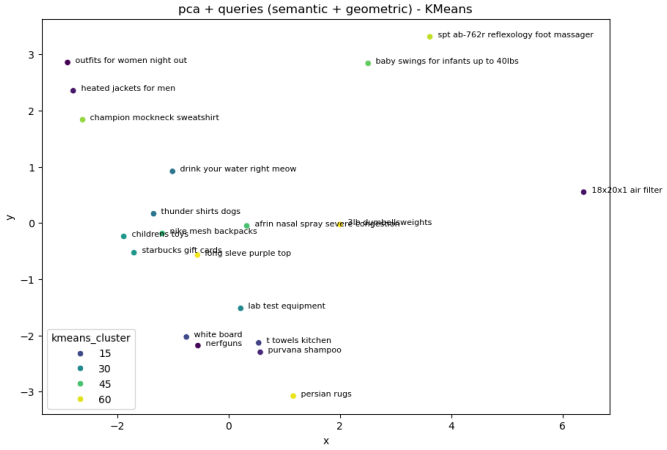


Fig. 12. A representation of queries using PCA - KMeans – geometric + semantic case

Cluster 23 (cloths and accessories + medium queries) : ‘hot pink berets for women’, ‘navy blue polo vest for men’,‘lepord print for women tops’ , ‘wrap dresses for women cap sleeve’,

Cluster 20 : (cloths and accessories + very big queries) : ‘I would just like to have a jacket with pockets, and is light and plain looking; no flashy colors.’, ‘The jacket I am planning on buying is a hoodie with high visibility reflective strips on it, for safety.’

Overall, this experiment shows that combining geometric and semantic features does not always improve clustering. In this case, it reduces the quality of clustering according to all metrics , but in terms of observing we could find some useful clusters.

VI. CONCLUSION

In this work, we studied clustering of Amazon real search queries in different cases : geometric features, semantic embeddings, and a combination of both , we used many clustering algorithms including KMeans, GMM, DBSCAN, and HDBSCAN , the results shows that geometric features lead to better separated clusters while semantic embeddings produce more meaningful clusters related to product categories and user intent . The hybrid approach combines both aspects but introduces more complexity and reduces internal evaluation scores .

And metrics such as Silhouette, Calinski–Harabasz, and Davies–Bouldin do not always fully reflect the semantic quality of clusters, especially when working with text embeddings [4] , In addition, using larger transformer models for embeddings or fine-tuning them on domain-specific data (e-commerce queries) could improve clustering quality and interpretability .

external evaluation methods based on user intent or labeled data could provide a more accurate measure of clustering performance.

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Github repo all code

https://github.com/ayoub-alla/clustering-queries-from-Amazon-ESCI-dataset